PAPER_663: UQFF Black Hole Inversion Probability

Subtitle: Derives the inversion criterion Θ_inv and Monte Carlo probability P_invert for UQFF-driven BH interior phase transitions. **Module:** UQFFBlackHoleInversion
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Abstract

We derive the UQFF Black Hole Inversion probability — the likelihood that a black hole's interior undergoes a [UA]/[SCm] gradient reversal, converting it to a white-hole-like state. The stochastic criterion Θ_inv > 1 yields P_invert ≈ 0.95 for Sgr A*.

1. Modified Schwarzschild Radius

$$r_{s,UQFF} = r_s \cdot (1 - \delta\rho), \quad \delta\rho = \frac{\rho_{SCm}}{\rho_{UA}} \approx 0.1$$

2. Inversion Energy

$$E_{inv,UQFF} = \frac{GM^2}{r_{s,UQFF}}$$

3. Inversion Probability Components

$$P_{inv} = f_{TRZ} \cdot \exp\left(-\frac{E_{inv}}{k_B T_H}\right)$$

$$\Phi_{inv} = \frac{1}{\delta\rho} \cdot \frac{GM}{c} \cdot (1 + f_{TRZ})$$

$$S_{U_m} = \exp\left(\frac{U_m}{k_B T_H}\right)$$

4. Combined Criterion

$$\Theta_{inv} = P_{inv} \cdot \Phi_{inv} \cdot S_{U_m}$$

Inversion occurs when Θ_inv > 1.

5. Stochastic Distribution

δρ, f_TRZ, U_m sampled from Gaussians → Θ_inv log-normal. P_invert = P(Θ_inv > 1) via Monte Carlo.

Numerical (Sgr A*): Θ_inv ≈ 2.7 → P_invert ≈ 0.95.

6. C++ Module

`UQFFBlackHoleInversion.h / .cpp` — Session 172 CP4 #247 — `UQFFBlackHoleInversionCalculator`

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